JEWELRY
TRADER

PHASE 1 · TEST SERVER GUIDE · 03 NOVEMBER 2026

Walking the system through a working day

What has been built, and the order to test it in — from an empty configuration screen to a completed sale, a refund and a commission figure.

Configure

Nine settings screens, in the order they depend on each other.

Catalogue

Photographs, fields, stones and a priced record.

Sell

The register, the storefront, refunds and commission.

Account

Every change logged, every override reasoned.

SECTION 01

What you are looking at

A point of sale, an inventory system and a shop, built for a dealer in estate and fine jewelry across several locations.

Phase 1 is complete and installed on the test server. Everything described in this document works now — there is no screen in here that is a mock-up and no number that was typed in by hand to make a demonstration read well.

The four things it does

Catalogues a piece
Photographs, attributes, stones and condition, with the required fields enforced before anything can be submitted.

Prices it
The craftsman's formula on your own rate table, every step shown and accountable.

Sells it
At the counter or on the website, through one payment path rather than two.

Accounts for it
Commission, refunds, overrides and an audit record no role can switch off.

What is deliberately not here

The reading service that looks at a photograph and proposes attributes, the description writer, automatic pricing and natural-language search are Phase 2. The seams for all four exist — the interfaces, the tables and the switches are in place — but no provider is connected and nothing calls out anywhere. The same is true of rentals, salesperson storefronts and the quarterly audit: the switches are visible in Settings, all off.

> If a screen says a feature is not active in Phase 1, that is the system telling you the truth rather than hiding something that half works.

How to use this guide

The sections are in dependency order. Configuration first, because everything else reads from it; then a piece through intake; then a sale. Working through it in order takes about ninety minutes and exercises every part of the system.

02	Signing in and the six roles	Two-factor enrolment
03	Configuration, in order	Nine screens
04	Your first item	Photographs to submission
05	Review, approve and list	Two deliberate steps
06	The storefront	What a customer sees
07	A sale at the register	Card, cash and split
08	Returns and exchanges	Refund rules
09	Commission, overrides and the audit log	After the sale
10	What the database holds	And what you add
11	What to tell us	Reporting back

SECTION 02

Signing in, and who can do what

Two-factor enrolment happens on first sign-in. Have an authenticator app open before you start.

First sign-in

1. **Open the admin panel**

   Go to `/admin`. You will be sent to the sign-in page. Use the Super Admin account created during installation.

2. **Enrol two-factor authentication**

   A QR code appears. Scan it with Google Authenticator, Authy, 1Password or any TOTP app, then enter the six-digit code. This is required, not optional, for the roles configured in Settings.

3. **Save the recovery codes**

   A set of single-use recovery codes is shown once. Save them somewhere other than the device running the authenticator. They are the only way back in if the phone is lost.

4. **You land on the dashboard**

   The sidebar is grouped into Operations, Administration, Reporting and Phase 2. What you can see depends on your role — a Super Admin sees everything.

> **Before go-live.** The step-up challenge symbols and answers that ship with the system are defaults and are published in the source repository. Replace them in `Settings → Security` before the system holds anything real. Until you do, the challenge is theatre.

The six roles

Permissions are granular and checked on the server, not just hidden in the interface. Hiding a button is a courtesy; the policy layer is the control.

ROLE	CAN	CANNOT
Super Admin	Everything, including settings and overrides	Nothing — but every action is still logged
Store Manager	Inventory, orders, discounts and returns above the staff threshold, reports for their locations	Change system settings or payment keys
Sales Staff	The register, create and edit inventory subject to approval, see their own commission	Approve their own items, discount beyond the threshold
Inventory Specialist	Create, edit and approve inventory for listing	Sell at the register
Accountant	Financial and commission reports, payment and refund records	Edit inventory or make sales
Customer Service	View orders and customers, process returns	Discount or change pricing

Custom roles can be built from the same permission set in `Roles & Permissions` — no code change needed.

> Worth testing: sign in as a Sales Staff user at one location and confirm you cannot see another location's inventory. That restriction is enforced in the data layer, not by hiding a menu.

SECTION 03

Configuration, in the order it depends on itself

Nine screens. Later ones read from earlier ones, so working down the list avoids doubling back.

Almost nothing in this system is fixed in code. Tax rates, commission structures, required fields, the rate table, which payment provider is live, which stones must carry a disclosure — all of it is a form in the admin panel. If you find something you want to change and cannot, tell us: that is a gap, not a limitation.

1. **General —** `Settings → General`

   Company name, logo, timezone, currency, receipt footer and business hours. The receipt footer appears on every printed and emailed receipt, so put the return policy there now and you will see it in Section 07.

2. **Locations —** `Settings → Locations`

   Four locations are installed as a starting point: Madison Ave, Workshop, Greenwich and Web. Rename them, correct the addresses, and set the tax rate for each one — the register takes its tax from the location serving the sale, so this has to be right before any selling test.

3. **Payments —** `Settings → Payments`

   Leave **test mode on**. Enter the Stripe publishable key, secret key and webhook signing secret from your Stripe test dashboard. Once saved they are encrypted and never shown again — the form displays a masked placeholder, and entering nothing leaves the stored key untouched. Switch on the payment methods you accept: card, cash, split.

4. **Pricing —** `Settings → Pricing` **and** `Pricing Control`

   Your opening rate table is already loaded: metal rates, the diamond size bands, the fancy-cut rates, clarity, colour and cut adjustments, treatment discounts, maker multipliers and the insurance figures by category.

   `Pricing Control` shows all four pricing layers at once and prices a real example piece through whatever is switched on, so you can see what each layer is worth in dollars rather than being told it is enabled. It also exports the whole configuration to a file and loads one back — useful before a change nobody is sure about.

5. **Field rules —** `Settings → Field Rules`

   Which intake fields are required, which are optional and which do not apply, per category. A ring needs a size; a brooch does not. Changes take effect immediately. The required rule is enforced on the server, so it cannot be worked around by the browser.

6. **Quality —** `Settings → Quality`

   The quality gauge and its checks. Each check can be ranked critical, standard, optional or disabled, and a failed critical check can be set to block listing.

7. **Security —** `Settings → Security`

   Which roles must use two-factor, session timeout, password policy, audit retention, and the step-up challenge symbols. Replace the default challenge answers here.

8. **Commission —** `Settings → Commission`

   The default commission structure — sales-based, profit-based, tiered or split — used when a salesperson has no individual arrangement. Individual agreements are assigned per person, with their written terms stored alongside the numbers, which California Labor Code §2751 requires.

9. **Feature flags —** `Settings → Feature Flags`

   Everything that is Phase 2, each with a line describing what it gates. All off. Leave them off — switching one on reveals a placeholder, not a feature.

> Every one of these changes writes an audit entry with the old and new values, including changes made by a Super Admin. Encrypted values are masked in the log, so rotating a Stripe key is recorded without the key being recorded.

SECTION 04

Your first item

Photographs, then fields, then stones, then a price. The target is under five minutes for a representative piece.

Go to `Inventory → Add / Edit Item`. The workflow is three steps and you can save a draft at any point.

1 **Upload the photographs**

Five to fifteen images. Tag each one by type — front, back, side, hallmark, gemstone, clasp, movement, signature, packaging, certificate. The first becomes the primary image unless you choose another. The count is guidance and never blocks submission.

2 **Fill the item record**

Every field carries a colour. 🔴 **Red** is required and blocks submission. 🟡 **Yellow** is a field the reading service will propose in Phase 2; today you type it. 🟢 **Green** is complete and valid. ⚪ **Grey** does not apply to this category. 🔵 **Blue** marks a value where a person replaced a suggestion.

Try leaving a red field empty and pressing Submit for review. It is refused by the server, not just by the browser.

3 **Record the stones**

The Stones section takes stone type, origin, shape, cut, weight, colour, clarity, cut grade, treatment and setting style, and holds as many stones as the piece has. The cut list includes the Old European and rose cuts, which carry higher rates than any modern cut in your table.

Two reference aids sit in this section's header: a hover tooltip explaining the 4Cs, and a **Reference chart** button that opens the jeweller's chart full size without leaving the form. Both are internal — neither appears on the shop.

4 **Price it**

Press **Suggest a price**. The craftsman's formula runs on the metal, the stones and the labour, and the result shows every step: determining factors, basic, wholesale, retail, then each pricing layer and what it added. The insurance value and negotiation floor come with it.

Accept the figures or type your own. Whatever you accept is stored with the arithmetic that produced it, so the price can still be explained in two years when every rate behind it has moved.

5 **Submit for review**

Once the red fields are complete, Submit for review moves the item to the queue and records the time. The elapsed time from creation to submission is tracked and shown to managers, so the five-minute target is measured rather than asserted.

Two things that will stop you, on purpose

WHAT HAPPENS	WHY
A ruby, sapphire or emerald with no treatment recorded cannot be listed	Corundum and beryl are treated as a matter of course. Silence is not the same as untreated, and nobody can derive the answer from a photograph — so the system refuses to publish until a person has said. "Unknown" is an acceptable answer; blank is not.
A metal the rate table does not price returns no price	Plated and filled marks — EPNS, Alpaca, gold-filled — are not in the table, so a piece entered as one of them will not price. That is the intended behaviour: it stops a plated piece being priced as solid.

SECTION 05

Review, approve and list

Approving an item and publishing it are separate actions, deliberately.

Sign in as a user with approval rights and open `Review Queue`.

1. **Open the submitted item**

   Everything is editable here. A reviewer can correct anything before approving.

2. **Approve**

   The item becomes **approved**. It is now a complete record — but it is not for sale and it is not on the website.

3. **List**

   A separate action publishes it. An item can be approved for the record without being offered for sale: a piece going to a client on approval, or one held back for a show, is approved but not listed.

> Listing is also where the stone disclosure check runs. If a ruby's treatment is missing, the item approves fine and refuses to list, with the reason on screen.

The quality gauge

Open `Inventory → [item] → Quality`. Sixty-one checks across nine stages, most of them derived from the record rather than asked of a person. The gauge is for your staff — it tells them what is missing. It is not shown to customers, and there is no score or star rating anywhere on the shop.

What customers see instead is a short panel of specific verified facts, each one attributed to the role and date that verified it. "Metal tested and confirmed 18k, appraiser, 14 October" is a claim that can be stood behind. "Quality: 5 stars" is not.

SECTION 06

The storefront

Public, keyword search, and strict about what it will show.

Open the site root in a private browsing window, so you are testing as a customer rather than as yourself.

- [ ] Browse the collection and filter by category, period, metal and price
- [ ] Search by keyword — this is a database search in Phase 1, not a semantic one
- [ ] Open a piece: gallery, description, the verified-facts panel, the stone detail
- [ ] Add to the bag and go through checkout with a Stripe test card
- [ ] Register an account and check the order appears in the history

What must never appear

Only items that are **listed** and whose stock is **in stock** are public. Draft, pending, approved-but-unlisted, sold, reserved and locked items are invisible, and are not reachable by guessing the URL either. This is worth testing directly: take the reference number of a draft item and try to open it on the shop. It returns a 404.

Disclosure

If a piece contains a lab-grown or simulated stone, the shop page says so. That line is built from the stone record each time the page is drawn, not written into the description — so no edit can remove it and no hand-written description can forget it. The only way to take the label off is to change what the stone is.

> Test cards, in test mode: `4242 4242 4242 4242` succeeds, `4000 0000 0000 9995` is declined for insufficient funds, `4000 0025 0000 3155` requires authentication. Any future expiry date and any CVC.

SECTION 07

A sale at the register

Full screen, no sidebar, large targets. The target is a complete sale in under two minutes.

Open `/pos`.

1. **Find the item**

   Search by reference, barcode or title. Results show the photograph, the price and whether it is in stock. Add it to the cart.

2. **Check the cart**

   Line items, running subtotal, and tax taken from the location's configured rate. Apply a discount as a percentage or a fixed amount.

3. **Try a discount above the threshold**

   Above the percentage set in settings, the discount needs a role that holds the approval permission. A Sales Staff user is stopped; a manager is not. Worth testing both ways.

4. **Take payment**

   Card, cash, or split across both on one order. Cash is recorded without touching the gateway. Card runs a real Stripe test charge.

5. **The receipt**

   On screen, printable on 80mm thermal stock, and emailed when the customer has an address on file. Your receipt footer from Section 03 appears here.

The thing to test hardest

Every piece is one of a kind, so two staff must not be able to sell the same item. When an order is paid, the stock flips to sold inside the same database transaction as the payment and the order status — there is no window in which a sold item still reads as available. If you can get two registers to sell the same reference, that is the most valuable bug you could find, and we would want to know immediately.

SECTION 08

Returns and exchanges

Look up the order, choose the items, choose how the money goes back.

Open `/pos/returns` and find the order by its number.

- [ ] Select the items coming back
- [ ] Choose the refund method — a card refund follows Stripe's rules; cash is unrestricted
- [ ] Optionally exchange for a different piece: the new line and the refund adjustment happen in one transaction
- [ ] Confirm the returned item goes back to in-stock and reappears on the shop
- [ ] Confirm the order status changes to refunded or partially refunded

Refunds run through the same payment path as the original sale, so a card refund is a real Stripe test refund and appears in your Stripe dashboard against the original charge.

SECTION 09

After the sale

Commission, overrides, locks and the record of what happened.

Commission

Calculated by a background job when an order is marked paid, never during checkout — so the arithmetic can never delay a customer at the counter. Open `Commission Report` within a minute of a test sale and the figure should be there. If it is not, the queue scheduler on the server is not running, which is the first thing to check.

Reports filter by staff and location, run daily, weekly or monthly, and export to CSV. The payroll export column mapping is configurable, because the payroll system this eventually feeds has not been chosen yet.

There are no automatic clawback deductions from commission, by design. California Labor Code §221 governs deductions from wages, so any reduction goes through the override system instead — reasoned, approved and logged.

Overrides and locks

`Override & Lock`. Any staff member can request an override; it takes effect only after someone with approval rights approves it, and a written reason is required in every case. A Super Admin overriding something still provides a reason and is still logged — there is no silent path.

Five lock types, each blocking exactly its own restricted actions:

LOCK	EFFECT
Full	Frozen — no sale, no edit, hidden from the shop
Sales	Cannot be sold or carted; can still be edited, re-photographed, re-appraised
Rental	Cannot be rented — relevant in Phase 2, present now so no migration is needed later
Edit	Cannot be edited, for a piece mid-dispute; still viewable and sellable
View	Hidden from the shop and from register search, for a legal hold or investigation

These are enforced in the service layer, so the restriction holds whether the request comes from the register, the website or anywhere else.

The audit log

`Audit Log`. Every create, update and delete on products, orders, payments, settings and users, with who, what, when, the before and after values, and the IP address. Sign-ins, two-factor passes and failures, and override actions are logged explicitly. No role is exempt, Super Admin included.

Retention is configurable. Note that financial records generally need longer retention than general activity — seven years for IRS recordkeeping — so set the retention period with that in mind rather than leaving it at the default.

SECTION 10

What is in the database, and what you add

The system is configured but has no trading data. That is deliberate.

Reference data that makes the application work is installed. Anything that is a record of your actual business is left empty for you to enter. Nothing below needs a developer — it is all entered through the admin panel.

Already there, and editable

WHAT	HOW MUCH	WHERE TO CHANGE IT
Roles and permissions	6 roles, 38 permissions	Roles & Permissions
Settings	122 values at their defaults	The nine settings screens
Locations	4 starter locations	Settings → Locations
Categories	Rings, necklaces, bracelets, brooches, earrings, watches	Settings → Field Rules
Field colour rules	31 rules	Settings → Field Rules
Rate table	Your opening table, fully loaded	Settings → Pricing
Payment gateway	The Stripe row	Settings → Payments
Step-up challenges	3 defaults — **replace these**	Settings → Security

Empty — you fill these

WHAT	HOW IT GETS FILLED
Staff accounts	The first was made at installation; add the rest in Roles & Permissions
Products, images, stones	Inventory → Add / Edit Item
Prices and stock	Written automatically as you enter a price and choose a location
Customers	The Customers screen, or captured at the register
Orders and payments	A completed sale, at the register or on the website
Commission plans and agreements	Settings → Commission, then assigned per person
Transfers, overrides, locks	Created as situations call for them

Fills itself

The audit log, the quality check records, commission figures, and the log of every live-rate call. Nothing to enter.

Stays empty in Phase 1

The tables behind the reading service, rentals, salesperson storefronts and the quarterly audit exist but hold nothing. They are there so that turning a feature on later is a switch rather than a database migration. An empty table here means the system is working as intended, not that something failed to install.

SECTION 11

What to tell us

What is useful to hear back, and what is already known.

Most useful

- Anything you expected to be able to change from a screen and could not
- A number that does not match what you would have worked out by hand — with the reference, so we can trace the arithmetic
- Any route by which a draft, sold or locked item becomes visible to a customer
- Any point where the register slows a sale down
- Wording that reads wrong to a jeweller — we have built the system, not spent thirty years in the trade

Already known, no need to report

- The live metals feed is off until a provider and key are entered
- The reading service, the description writer and semantic search are Phase 2 and do not run
- Phase 2 nav items show a placeholder — that is what they are for
- Plated and filled metals do not price; that is the hard stop working

Three differences to settle

When you compare the system's figures against your own worked examples, three differences will show up. They are in the documents rather than in the code, and we would like your decision on the first two.

- **Rounding.** The system rounds up to the nearest $0.50, as specified. Your appraisal records round to the nearest $5 or $25. The increment is a single setting — tell us the number you want.
- **Insurance multiplier.** Your records use ×1.35 as standard. Your signed rate table has men's accessories at ×1.30. The system uses the table.
- **The ×6.0 note.** Appraisal record #2 states that the ceiling triggered a review, but its multipliers compound to ×1.518 — nowhere near it. Worth correcting in that document, as it reads as a statement about what the system did.